End-to-End Digital Customer Onboarding System with Risk-Based Decisioning and Real-Time Analytics

Business Analysis Case Study | Process Optimization | KYC Automation

Project Summary

This project presents the design of an end-to-end digital onboarding system aimed at transforming a manual, time-intensive process into an automated, risk-driven workflow. The solution leverages automated document verification, risk-based decisioning, and real-time dashboards to reduce onboarding time from 5–7 days to under 24 hours while improving operational efficiency and customer experience.

1. Project Background

Organizations currently rely on manual, branch-based onboarding processes that take 5–7 business days, resulting in 30% customer drop-off, high administrative overhead, and frequent data errors. This project designs a digital onboarding system with automated verification, risk-based approval workflows, and real-time analytics to reduce onboarding time to under 24 hours.

2. Project Objectives

- Reduce onboarding time from 5-7 days to under 24 hours
- Reduce manual data entry errors by 80%
- Achieve 85%+ application approval rate
- Provide real-time visibility into onboarding pipeline via dashboard
- Ensure compliance with document verification standards
- Improve customer experience by enabling real-time application tracking and eliminating the need for physical branch visits

3. Success Metrics (KPIs)

- **Onboarding Time:** Reduced from 5–7 days to under 24 hours
- **Approval Rate:** ≥ 85% for valid applications
- **Error Reduction:** 80% decrease in manual data entry errors
- **SLA Compliance:** ≥ 95% applications processed within SLA
- **Customer Drop-off Rate:** Reduced by 30%

4. Stakeholder and Pain Points

4.1 Stakeholders

- **Customer –** Submits application and track status
- **Onboarding Officer –** Reviews and processes applications
- **Compliance Team –** Handles high-risk and flagged cases
- **Admin –** Monitors system operations
- **Management –** Tracks KPIs and performance

4.2 Pain Points by Stakeholder

4.2.1 Customer:

- Required to visit branch physically
- No visibility into application status
- Long processing time leads to abandonment

4.2.2 Onboarding Officer:

- Manual document verification is time-consuming
- No centralized system (email/spreadsheets)
- Difficult to track pending applications

4.2.3 Compliance Team

- No automated risk identification
- Manual review of all applications
- Limited audit traceability

4.2.4 Management

- No real-time dashboard
- Cannot identify bottlenecks
- Reports are manual and delayed

5. Project Scope

5.1 In Scope:

- Online customer registration
- Document upload and validation
- Multi-stage approval workflow
- Risk-based decision routing

- Account activation
- Notifications (email/SMS)
- Admin and officer dashboards
- Reporting and analytics

5.2 Out of Scope:

- Mobile app development
- Payment processing
- Third-party credit bureau integration
- Legacy system migration

6. Assumptions and Constraints

6.1 Assumptions:

- Customers have access to internet and a device
- Documents submitted are in PDF or image format
- Officers will be trained before system goes live
- Compliance rules are predefined and documented

6.2 Constraints:

- Project is a portfolio simulation — no real development
- No actual third-party API integrations

7. Problem Statement

Traditional onboarding is slow, manual, and inefficient. Customers wait 5–7 business days for account activation, leading to ~30% drop-off and high operational overhead. Even in digitized systems, applications are processed using a uniform workflow regardless of risk level, resulting in unnecessary manual reviews and inefficient resource utilization.

Organizations also lack real-time visibility into their onboarding pipeline. Without centralized tracking and analytics, they cannot identify bottlenecks, monitor SLA compliance, or optimize operational performance.

This project addresses these challenges by designing an end-to-end digital onboarding system with automated document validation, risk-based workflow routing, and real-time dashboards—reducing onboarding time to under 24 hours while improving efficiency, compliance, and customer experience.

8. Project Execution Timeline (6 Weeks)

Week	Focus	Deliverables
Week 1	Business Analysis & Problem Definition	Defined project scope, stakeholder analysis, and documented business problem and objectives
Week 2	Requirements Gathering & Documentation	Prepared BRD including assumptions, constraints, and high-level system requirements
Week 3	Functional Design	Developed FRD, user stories, and detailed use cases to define system behavior
Week 4	Process & System Design	Created AS-IS and TO-BE process flows, system architecture, and wireframes for key screens
Week 5	Data & Analytics Design	Designed database structure, SQL queries, and built Power BI dashboard for reporting and insights
Week 6	Final Delivery & Presentation	Compiled portfolio documentation, created system walkthrough, and finalized project presentation

9. Current Process Analysis (AS-IS)

The AS-IS analysis evaluates the current onboarding workflow to identify inefficiencies, delays, and process gaps across customer, operational, and compliance stages.

Key issues include:

• High dependency on manual document verification

• No risk-based prioritization of applications

• Lack of centralized tracking and visibility

• Multiple delays due to repeated communication cycles

• Limited audit and performance monitoring

These inefficiencies result in extended onboarding timelines (5–7 days), increased operational workload, and higher customer drop-off rates.

Steps 6–7 represent a rework cycle caused by missing validation and manual communication, increasing overall onboarding time.

Table 1 – AS-IS Onboarding Process Flow and Pain Point Analysis

Step	Actor	Process Step Description	Time	Pain Point
1	Customer	Initiates application via branch or onboarding portal	30–60 min	Requires physical visit or lacks a clear digital entry point
2	Customer	Fills application form with personal and contact details	15–30 min	Form complexity leads to errors or incomplete fields
3	Customer	Submits identity and address proof documents	10–15 min	No clear checklist → incorrect/missing documents
4	Onboarding Officer	Reviews submitted documents manually	30–60 min	Manual document review leads to inconsistent validation outcomes
5	Onboarding Officer	Validates application completeness and accuracy	30–45 min	Lack of system validation increases dependency on manual checks
6	Onboarding Officer	Requests additional documents or corrections	1–2 days	Manual communication causes delays
7	Customer	Resubmits corrected documents	1–3 days	Rework cycle increases turnaround time
8	Onboarding Officer	Enters customer data into internal system	30–45 min	Manual data entry introduces duplication and accuracy issues

9	Onboarding Officer	Routes application to compliance team	1 day	Lack of centralized workflow leads to poor tracking and visibility
10	Compliance Team	Reviews application and verifies documents	1–2 days	All applications are reviewed uniformly without risk-based prioritization
11	Compliance Team	Requests additional checks or clarification if required	1–2 days	Further delays due to repeated verification cycles
12	Compliance Team	Makes approval or rejection decision	2–4 hr	No structured decision tracking or audit visibility
13	Onboarding Officer	Communicates rejection to customer	1 day	No self-service correction results in process restart
14	Admin	Creates customer account in system	4–8 hr	Manual processing is slow and error-prone
15	System	Sends notification to customer (email/SMS)	1 day	No real-time status tracking for customer

Process Summary

Total Estimated Onboarding Time: *5–7 business days (including rework cycles)*

Key Observations

- The onboarding process is heavily dependent on manual review and verification, leading to delays across multiple stages
- Rework cycles due to missing or incorrect documents significantly increase overall processing time

- All applications are processed uniformly without risk-based prioritization, resulting in inefficient use of compliance resources
- Lack of a centralized workflow system creates challenges in tracking application status and identifying bottlenecks
- Manual data entry introduces risk of errors and duplication, requiring additional validation and correction
- Customers have no real-time visibility into application status, leading to poor experience and higher drop-off rates
- The process contains multiple non-value-adding steps, particularly in manual validation, data entry, and rework cycles, which significantly impact efficiency and scalability

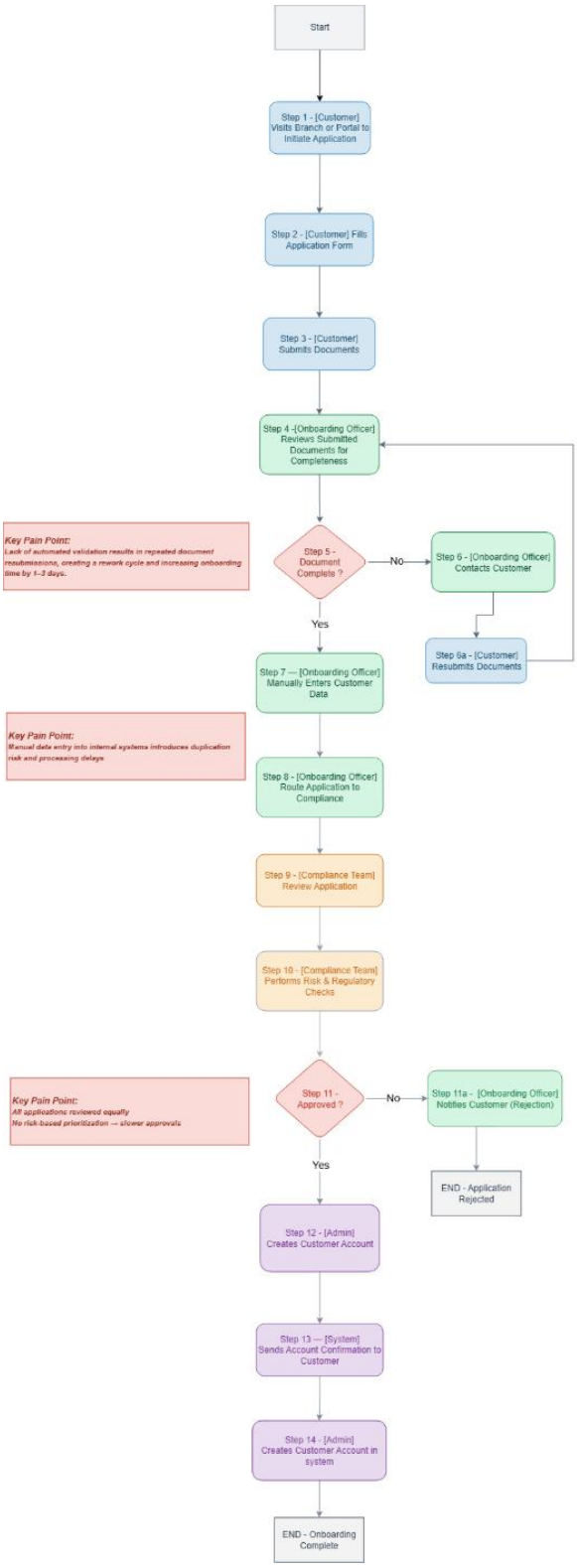
Operational Impact

- High customer drop-off due to prolonged onboarding timelines
- Increased administrative workload for onboarding and compliance teams
- Inefficient allocation of resources due to lack of prioritization
- Limited ability for management to monitor performance and identify process inefficiencies
- Delayed decision-making due to lack of real-time reporting and operational insights

Figure 1 – AS-IS Onboarding Process Flow

This diagram illustrates the current manual onboarding process across customer, onboarding, and compliance stages. It highlights key inefficiencies such as manual validation, repeated rework cycles, lack of risk-based prioritization, and limited system integration, resulting in processing delays and poor operational visibility.

These gaps form the foundation for designing an optimized TO-BE onboarding solution focused on automation, risk-based decisioning, and real-time visibility.


Start
Step 1 - [Customer] Visits Branch or Portal to Initiate Application
Step 2 - [Customer] Fills Application Form
Step 3 - [Customer] Submits Documents
Step 4 -[Onboarding Officer] Reviews Submitted Documents for Completeness
Key Pain Point:
Lack of automated validation results in repeated document resubmissions, creating a rework cycle and increasing onboarding time by 1–3 days.
Step 5 - Document Complete ?
No
Step 6 - [Onboarding Officer] Contacts Customer
Step 6a - [Customer] Resubmits Documents
Yes
Step 7 — [Onboarding Officer] Manually Enters Customer Data
Key Pain Point:
Manual data entry into internal systems introduces duplication risk and processing delays
Step 8 - [Onboarding Officer] Route Application to Compliance
Step 9 - [Compliance Team] Review Application
Step 10 - [Compliance Team] Performs Risk & Regulatory Checks
Key Pain Point:
All applications reviewed equally
No risk-based prioritization → slower approvals
Step 11 - Approved ?
No
Step 11a - [Onboarding Officer] Notifies Customer (Rejection)
END - Application Rejected
Yes
Step 12 - [Admin] Creates Customer Account
Step 13 — [System] Sends Account Confirmation to Customer
Step 14 - [Admin] Creates Customer Account in system
END - Onboarding Complete

10. Future State Process Design (TO-BE)

10.1 Overview

The TO-BE process introduces automation, intelligent decisioning, and real-time validation to eliminate inefficiencies identified in the AS-IS workflow.

The redesigned flow reduces onboarding time from days to minutes/hours by minimizing manual intervention and enabling risk-based processing.

10.2 Process Flow

The table below outlines the redesigned TO-BE onboarding process, highlighting system components, automation points, and improvements over the AS-IS workflow.

Table 2 – TO-BE Onboarding Process and System Design

Step	Actor	System Component	Process Step Description	Time	Improvement
1	Customer	Onboarding Portal	Accesses digital onboarding portal from any device	2–5 min	Eliminates physical branch visit
2	Customer	Onboarding Portal	Fills smart application form with real-time field validation and error prompts	10–15 min	Prevents incomplete/incorrect submissions
3	Customer	Document Upload Service	Uploads identity and address documents via guided checklist	5–10 min	Ensures correct documents upfront
4	System	Document Validation Service	Validates document format, size,	Instant	Replaces manual verification (30–60 min)

			and completeness, and extracts key data using OCR		
5	System	Notification Service	If validation fails, sends automated alert specifying the issue (e.g., missing page, unclear image)	Instant	Eliminates 1–2day follow-up delay
6	Customer	Onboarding Portal	Resubmits corrected documents via portal	15–30 min	Enables fully digital correction without physical visit
7	System	Data Extraction Engine	Automatically populates customer details into system using extracted document data	Instant	Eliminates manual data entry errors
8	System	KYC Verification Service	Performs automated identity verification via external services (government/third-party)	1–3 min (API dependent)	Replaces manual KYC checks
9	System	KYC Verification Service	If verification fails, retries automatically and routes to	Additional 5 min	Prevents process failure and ensures continuity

			manual review if unresolved		
10	System	Risk Scoring Engine	Evaluates application using identity match results, data consistency checks, duplicate detection, and predefined risk indicators	Instant	Enables intelligent risk-based decisioning
11	System	Decision Engine	Evaluates risk level and routes application	Instant	Enables automated decision routing
12	System	Decision Engine	Low-risk applications are auto-approved	Instant	Enables fully automated approval for low-risk applications
13	Onboarding Officer	Review Dashboard	Medium-risk applications are routed for manual review	2–4 hr	Ensures manual review is limited to medium-risk applications only
14	Compliance Team	Compliance System	High-risk applications are escalated for compliance review	4–8 hr	Ensures compliance review is focused on high-risk cases only
15	System	Audit Logging Service	Logs all actions (validation, scoring, decisions) with timestamp,	Instant	Enables full traceability for compliance and regulatory audits

			actor ID, and reason		
16	System	Notification Service	If rejected, sends notification with reason and resubmission option	Instant	Self-service correction
17	System	Core Banking System	Creates account automatically upon approval	Instant	Eliminates manual account setup
18	System	Notification Service	Sends account confirmation via email/SMS with login details	Instant	Real-time communication

10.3 TO-BE Process Summary

• End-to-end onboarding completed within minutes to hours
• Manual intervention limited to medium/high-risk cases only
• Automated validation eliminates rework cycles
• Real-time notifications improve customer experience
• Full audit logging ensures compliance and traceability

10.4 AS-IS vs TO-BE Comparison

• Onboarding Time: 5–7 days → < 24 hours
• Manual Steps: High → Minimal
• Error Rate: High → Reduced by 80%
• Customer Visibility: None → Real-time tracking
• Resource Utilization: Inefficient → Optimized via risk-based routing

The TO-BE design directly addresses the inefficiencies identified in the AS-IS process, enabling a streamlined, scalable, and automated onboarding experience.

11. Business Rules and Decision Logic

11.1 Business Rules

11.1.1 Rule 1 – Document Validation

Trigger	Customer uploads identity and address documents via portal
Condition	System checks file format (PDF/JPG/PNG), size(<5MB), completeness, and readability via OCR
Action – Pass	Documents valid: proceed to Data Extraction Engine
Action - Fail	Documents invalid: Notification Service sends automated alert with exact reason
Exception	If OCR fails or documents unreadable: marked invalid: customer prompted to resubmit

11.1.2 Rule 2 – Data Consistency Check

Trigger	After document data extraction is complete
Condition	Extracted data (name, DOB, ID number) must match user-entered form data
Action – Pass	Data matches: proceed to KYC verification
Action - Fail	Mismatch detected: flagged as Medium or High risk depending on severity
Exception	If data extraction fails: routed to manual review queue with officer alert

11.1.3 Rule 3 – KYC Verification

Trigger	After successful document verification
Condition	Identity verification via external KYC API
Action – Pass	Verification successful: proceed to Risk Scoring Engine
Action - Fail	Verification failed: marked as High Risk or rejected on failure reason
Exception	If API fails: retry 2 times: if still fails: route to manual review queue with alert

11.1.4 Rule 4 – Risk Scoring Logic

Trigger	After successful KYC verification
Condition	System evaluates identity match, data consistency, duplicate detection, and external risk signals
Action – Pass	Score 0-40: Auto approved

Action - Fail	Score 41-70: Officer review Score 71-100: Escalated to Compliance Team
Exception	If scoring engine fails route to manual review + notify supervisor

11.2 Risk Scoring Model

11.2.1 Scoring Overview

The Risk Scoring Engine evaluates 4 weighted factors that together produce a total score between 0 and 100.

Weights are assigned based on the severity and regulatory impact – identity verification and duplicate detection contribute higher scores due to their strong association with fraud.

Total Risk Score is calculated as:	
+ Factor 1 – Identity Verification	*[Max 30 points]*
+ Factor 2 – Data Consistency	*[Max 25 points]*
+ Factor 3 – Duplicate Detection	*[Max 25 points]*
+ Factor 4 – External & Behavioral Risk Signals	*[Max 20 points]*
Maximum Possible Score = 100	

Each factor evaluates distinct dimensions to avoid double-counting risk signals.

11.2.2 Factor 1 – Identity Verification (Max 30 points)

What it checks: - Whether the person is who they claim to be based exclusively on KYC API results and government database verification.

Note: This factor covers identity confirmation only. Cross-document data validation is handled separately in Factor 2.

Check Performed	Risk Points	Justification
KYC API verified successfully	0	No identity risk
Name on ID does not match government record	+10	KYC verification failure
DOB does not match government record	+10	Identity inconsistency
ID number not found in government database	+20	High fraud indicator
Photo on ID does not match selfie	+15	Identity fraud risk

11.2.3 Factor 2 – Data Consistency (Max 25 points)

What it checks: - Whether data across all submitted documents is internally consistent and valid.

Note: This factor covers cross-document validation only. Identity API verification is handled separately in Factor 1

Check Performed	Risk Points	Justification
All data fields consistent across documents	0	No consistency risk
Address mismatch between ID and proof of address	+10	Inconsistent records across submitted documents
Name spelling variation across submitted documents	+5	Minor inconsistency – may indicate error or alias
Expired document submitted	+15	Invalid identity proof – document no longer legally valid
Document issued within 30 days of application date	+10	Recently issued document is a known fraud indicator
Document issued more than 10 years ago	+5	May indicate outdated or unrenewed identity document

11.2.4 Factor 3 – Duplicate Detection (Max 25 points)

What it checks: - Whether the applicant already exists in the system under the same or different application.

Check Performed	Risk Points	Justification
No existing record found	0	Clean new applicant
Same email already registered	+10	Possible duplicate account
Same phone number already registered	+10	Possible duplicate account
Same national ID already registered	+20	Strong fraud indicator
Previous application rejected in last 30 days	+15	Recent rejection history
Multiple applications submitted in short window	+25	High fraud risk

11.2.5 Factor 4 – External & Behavioral Risk Signals (Max 20 points)

What it checks: - Known external risk flags from regulatory watchlists and behavioral anomalies that indicate elevated risk.

Note: IP tracking, device flags, and submission time patterns are simplified assumptions used to simulate real-world fraud detection signals in this project.

Check Performed	Risk Points	Justification
No existing record found	0	Clean risk profile
High-risk geographic location	+10	Regional risk profile
Politically Exposed Person (PEP) Flag	+15	Regulatory requirement
Sanctions watchlist match	+20	Immediate compliance flag
Unusual submission time (e.g. 3am)	+5	Behavioral anomaly
Device or IP flagged previously	+10	Technical risk signal

11.3 Scoring Decision Table

Total Score	Risk Level	Routing Decision	Review Owner	SLA Target
0 – 40	Low	Auto approved by Decision Engine	System	Under 1 hour
41 – 70	Medium	Routed to Officer Review Queue	Onboarding Officer	Under 4 hours
71 – 100	High	Escalated to Compliance Team	Compliance Team	Under 24 hours

Boundary rule: Scores of exactly 40 follow the Low threshold. Scores of exactly 70 follow the medium threshold. This is documented to prevent decisioning ambiguity.

11.4 Real-World Scoring Examples

Example 1 – Low Risk (Score:15)

Scenario: New customer, all document match, no duplicates, clean profile, located in high-risk region

Factor	Check	Points
Identity Verification	KYC verified, all data matches government record	0
Data Consistency	All documents consistent	0
Duplicate Detection	No existing record found	0
External & Behavioral Signals	High-risk geographic location	+10
Total Score	**10**	
Decision	**Auto Approved**	

Example 2 – Medium Risk (Score:50)

Scenario: Name spelling variation found on ID, address mismatch between documents, same email already registered, application submitted from flagged IP.

Factor	Check	Points
Identity Verification	Name spelling variation on ID	+10
Data Consistency	Address mismatch + suspicious doc date	+20
Duplicate Detection	Same email already registered	+10
External & Behavioral Signals	Flagged IP	+10
Total Score:	**50**	
Decision:	**Officer Review**	

Example 3 – High Risk (Score:70)

Scenario: ID not found in government database, same national ID already exists, PEP flag triggered.

Factor	Check	Points
Identity Verification	ID not found in government database	+20
Data Consistency	Expired document submitted	+15
Duplicate Detection	Same national ID already registered	+20
External & Behavioral Signals	PEP flag triggered	+15
Total Score	**70**	
Decision	**Compliance Escalation**	

11.5 Exception Handling

Scenario	System Action
Scoring engine fails to load	Route to manual review queue immediately
Scoring engine times out	Retry once after 30 seconds
Score cannot be calculated	Flag application + notify supervisor
Partial score calculated	Treat as High Risk by default – escalate to compliance

11.6 Business Justification

Decision	Why
Auto-approve at 0-40	Conservative threshold ensures only clearly low-risk applications bypass manual review
Officer review at 41-70	Medium complexity cases benefit from human judgment

Compliance at 71-100	High-risk cases require regulatory scrutiny and full audit trail
Higher weight on identity + duplicates	Strongest fraud indicators based on regulatory impact
Manual fallback on engine failure	Ensures no application is lost or stuck due to system error

The defined business rules and scoring logic ensure consistent, transparent, and auditable decision-making across all onboarding scenarios.

11.7 Process Flow Integration (TO-BE Flow Explanation)

The TO-BE process flow diagram visually represents how the defined business rules and decision logic are executed within the onboarding system.

Each step in the flow is directly governed by the rules defined above:

• **Document Validation (Rule 1)** is applied immediately after document upload, ensuring only valid documents proceed further in the process.

• **Data Consistency Check (Rule 2)** ensures that extracted data aligns with user-provided inputs before KYC verification.

• **KYC Verification (Rule 3)** validates identity using external services, with retry and fallback mechanisms to ensure process continuity.

• **Risk Scoring Logic (Rule 4)** determines application routing based on calculated risk classification.

The flowchart highlights key decision points where the system dynamically routes applications:

• Document Validation Decision → Determines whether the application proceeds or requires correction

• KYC Verification Decision → Ensures identity validation before risk assessment

• Risk Classification (Low / Medium / High) → Drives automated approval or manual review routing

Based on the risk classification:

• Low-risk applications are automatically approved by the Decision Engine

• Medium-risk applications are routed to onboarding officers for review

• High-risk applications are escalated to the compliance team for further investigation

The flow also incorporates exception handling scenarios such as:

• Invalid documents triggering automated notifications and resubmission

• KYC API failures triggering retries and fallback to manual review

• Scoring engine failures routing applications to manual review with alerts

This integration ensures that the onboarding process is:

• Automated where possible

• Controlled through defined business rules

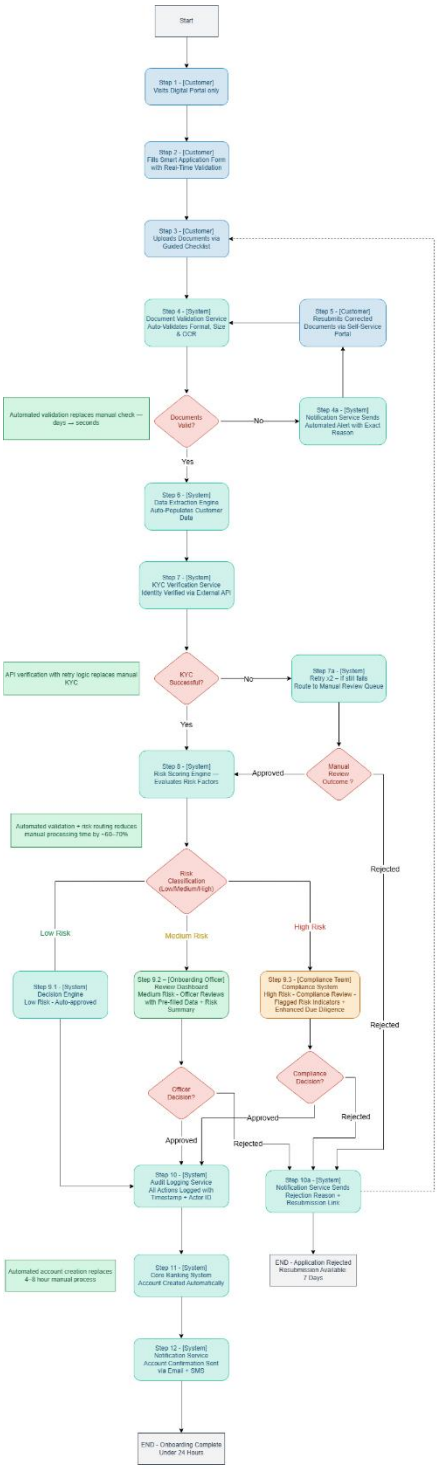
• Scalable through risk-based routing

• Fully traceable through audit logging

The result is a streamlined onboarding workflow that reduces processing time, minimizes manual intervention, and improves decision accuracy.

Figure 2 – TO-BE Onboarding Process Flow

This diagram represents the redesigned onboarding process, showcasing automated validation, risk-based decisioning, and dynamic application routing. It highlights key decision points such as document validation, KYC verification, and risk classification, enabling faster processing, reduced manual intervention, and improved operational efficiency.



LEGEND

- Customer Action
- Automated System Action
- Onboarding Officer Action
- Compliance Team Action
- Decision Point
- Improvement Annotation

12. Business Requirements Document (BRD)

12.1 Executive Overview

This Business Requirements Document (BRD) defines the functional and operational requirements for redesigning the customer onboarding process.

The current onboarding process is manual, branch-dependent, and inefficient, resulting in long processing times, high customer drop-off, and limited operational visibility.

The proposed solution introduces automation through document validation, API-based KYC verification, and risk-based decision routing. The objective is to reduce onboarding time from 5–7 days to under 24 hours while improving accuracy, scalability, and compliance.

12.2 Problem Statement

The current onboarding process relies heavily on manual verification and lacks automation, leading to delays of 5–7 business days.

Applications are processed uniformly without risk-based prioritization, increasing unnecessary workload for onboarding and compliance teams. Additionally, the absence of centralized tracking limits visibility into application status and operational performance.

Quantified Impact of Current State:

Problem	**Business Impact**
Manual branch-dependent process	30% customer drop-off rate
No automated document validation	1–2-day delays per application
Uniform compliance review for all applications	Inefficient use of compliance resources
Manual data entry into internal systems	High error and duplication rate
No real-time tracking or visibility	Zero management visibility into pipeline

12.3 Business Objectives

No	Objective	Target	Measurement
1	Reduce onboarding turnaround time	5-7 days – Under 24 hours	System timestamp logs
2	Minimize manual verification through automation	80% reduction in manual steps	Process audit
3	Implement API-based KYC with retry logic	99% verification coverage	KYC service logs

4	Introduce risk-based routing for applications	60-70% reduction in compliance workload	Routing analytics
5	Enable near real-time operational visibility	Dashboard refresh within 60 seconds	Dashboard monitoring
6	Achieve application approval rate of 85%+	85%+ approval rate	Decision engine reports
7	Ensure full compliance audit trail	100% action logging	Audit log coverage

12.4 Proposed Solution Overview

12.4.1 System Overview

The proposed Customer Onboarding System is a web-based digital platform that enables customers to complete the full onboarding process remotely without visiting a branch. The system automates document validation using OCR technology, performs identity verification via external KYC APIs, routes applications through a risk-based decision engine that assigns a risk score (0–100) and automatically determines application routing based on risk classification.

12.4.2 Process Improvements

• Automated document validation using OCR eliminates manual checks
• API-based KYC verification replaces manual identity validation
• Risk-based decision engine routes applications dynamically
• Real-time dashboards provide operational visibility

12.4.3 Decision Logic

Applications are processed based on risk score:
• Low Risk (0–40): Auto-approved instantly
• Medium Risk (41–70): Routed to onboarding officer
• High Risk (71–100): Escalated to compliance team

12.4.4 Exception Handling

The system shall handle exceptions such as API failures, incomplete data, and validation errors by routing applications to manual review with defined retry mechanisms. This ensures process continuity and prevents applications from being lost or stuck due to system failures.

12.4.5 Business Outcomes

The proposed system delivers measurable improvements in efficiency, accuracy, and customer experience. It reduces onboarding time from 5–7 days to under 24 hours, decreases manual review workload by 60–70%, and minimizes data errors through automation.

The system also provides real-time dashboards for operational visibility, ensures full audit traceability for compliance, and enhances customer experience through self-service capabilities and real-time status tracking.

12.5 Functional Requirements

12.5.1 Customer Registration & Application

Req ID	Functional Requirement	Priority	Stakeholder
FR-001	The system shall allow customers to register and initiate onboarding entirely online without visiting a branch	Must Have	Customer
FR-002	The system shall provide a smart application form with real-time inline field validation and error prompts	Must Have	Customer
FR-003	The system shall guide customers through document upload using a predefined checklist showing required documents	Must Have	Customer
FR-004	The system shall allow customers to save application progress and resume later	Should Have	Customer

12.5.2 Document Validation

Req ID	Functional Requirement	Priority	Stakeholder
FR-005	The system shall validate uploaded documents for format (PDF/JPG/PNG), file size (max 5MB), and completeness	Must Have	IT Team
FR-006	The system shall use OCR to extract key data fields from submitted documents automatically	Must Have	Onboarding Officer
FR-007	The system shall notify customers of validation errors in real time with clear reason and corrective action	Must Have	Customer

12.5.3 KYC Verification

Req ID	Functional Requirement	Priority	Stakeholder
FR-008	The system shall integrate with external government or third-party APIs to verify customer identity	Must Have	Compliance Team
FR-009	The system shall complete KYC verification within 1–3 minutes under normal operating conditions	Must Have	Customer
FR-010	The system shall retry KYC verification automatically up to 2 times in case of API failure	Must Have	IT Team
FR-011	The system shall route applications to manual review queue if KYC API fails after 2 retry attempts	Must Have	Onboarding Officer

12.5.4 Risk Scoring Engine

Req ID	Functional Requirement	Priority	Stakeholder
FR-012	The system shall assign a risk score between 0-100 based on 4 defined factors: identity verification, data consistency, duplicate detection, and external risk signals	Must Have	Compliance Team
FR-013	The system shall classify applications into Low (0-40), Medium (41-70), and High (71-100) risk categories	Must Have	Compliance Team
FR-014	The system shall route applications to manual review and notify supervisor if the scoring engine fails	Must Have	Management

12.5.5 Decision Routing

Req ID	Functional Requirement	Priority	Stakeholder
FR-015	The system shall automatically approve Low-risk applications without manual intervention	Must Have	Management
FR-016	The system shall route Medium-risk applications to onboarding officer review queue with pre-filled data and risk summary	Must Have	Onboarding Officer
FR-017	The system shall escalate High-risk applications to compliance team with flagged risk indicators and enhanced due diligence flag	Must Have	Compliance Team

12.5.6 Account Creation & Activation

Req ID	Functional Requirement	Priority	Stakeholder
FR-018	The system shall automatically create customer account upon approval without manual IT intervention	Must Have	IT Team
FR-019	The system shall send account confirmation with login details via email and SMS immediately upon activation	Must Have	Customer

12.5.7 Notifications & Communication

Req ID	Functional Requirement	Priority	Stakeholder
FR-020	The system shall send automated notifications to customers at every key stage of the onboarding process	Must Have	Customer
FR-021	The system shall send rejection notifications with specific reason and a resubmission link	Must Have	Customer
FR-022	The system shall allow customers to resubmit rejected applications via self-service portal within 7 days	Must Have	Customer
FR-023	The system shall permanently close applications after 3 failed resubmission attempts and notify customers	Must Have	Customer

12.5.8 Audit Logging & Compliance

Req ID	Functional Requirement	Priority	Stakeholder
FR-024	The system shall log every action – validation, scoring, routing, decision – with timestamp, actor ID, and reason	Must Have	Compliance Team
FR-025	The system shall maintain an immutable audit log accessible only to compliance team and management	Must Have	Compliance Team
FR-026	The system shall automatically escalate SLA breaches to supervisor and flag on admin dashboard	Should Have	Management

12.5.9 Reporting & Dashboard

Req ID	Functional Requirement	Priority	Stakeholder
FR-027	The system shall provide a real-time management dashboard showing onboarding volume, approval rates, and bottlenecks	Must Have	Management

FR-028	The system shall provide an officer dashboard showing pending review queue, SLA status, and application details	Must Have	Onboarding Officer
FR-029	The system shall provide a compliance dashboard showing high-risk escalations and audit trail	Must Have	Compliance Team
FR-030	The system shall generate compliance reports showing approval and rejection rates by risk level	Should Have	Management

12.6 Non-Functional Requirements

Req ID	Non-Functional Requirement	Category	Priority
NFR-001	The system shall process and complete low-risk applications within 1 hour of submission	Performance	Must Have
NFR-002	The system shall complete KYC verification within 1-3 minutes under normal operating conditions	Performance	Must Have
NFR-003	The system shall ensure accurate data extraction and validation with error handling for incomplete or inconsistent inputs	Data Quality	Must Have
NFR-004	The system shall maintain complete and immutable audit logs for all actions for regulatory compliance	Compliance	Must Have
NFR-005	The system shall ensure all customer data is encrypted in transit and at rest using industry-standard security protocols	Security	Must Have
NFR-006	The system shall support scalability to handle high application volumes without performance degradation	Scalability	Must Have
NFR-007	The system shall maintain 99.9% uptime excluding scheduled maintenance windows	Availability	Should Have
NFR-008	The system shall provide dashboard updates with a maximum 60 second data refresh rate	Performance	Should Have

12.7 Stakeholder Register

Stakeholder	Role	Responsibility	Interest	Influence	Expectation
New Customer	End User	Submits application and documents	Fast, transparent onboarding	Low	Complete onboarding <24 hrs with

					status tracking
Onboarding Officer	Reviewer	Reviews medium-risk applications	Efficient processing queue	High	Clear queue with SLA priority and pre-filled data
Compliance Team	Regulator	Reviews high-risk escalations	Regulatory compliance	High	Accurate risk flags with full audit trail
IT Team	Maintainer	Maintains portal and integrations	System stability	Medium	High availability and failure handling
Admin	Monitor	Tracks system operations	Operational visibility	Medium	Real-time dashboard for monitoring
Management	Decision Maker	Reviews KPIs and performance	Business outcomes	High	Real-time insights for decision-making

12.8 Expected Business Benefits

No.	Benefit	Impact
1	Onboarding time reduced from 5-7 days to under 24 hours	Faster customer activation and revenue generation
2	60-70% reduction in manual processing workload	Lower operational cost and better resource utilization
3	Improved data accuracy through automated validation and OCR, reducing manual errors and compliance risks	Fewer errors, corrections, and compliance risks
4	Risk-based routing focuses compliance on high-risk cases only	Better allocation of compliance resources
5	Real-time customer notifications reduce drop-off from ~30%	Improved customer experience and satisfaction
6	Full immutable audit trail enables regulatory compliance	Reduced compliance risk and audit exposure
7	Management dashboard enables data-driven decisions	Improved operational visibility and performance tracking

12.9 Success Metrics

Metric	Current State	Target	Measurement Method
Onboarding completion time	5-7 business days	Under 24 hours	System timestamp logs
Customer drop-off rate	~30%	Under 15%	Abandonment tracking
Document resubmission rate	~40%	Under 10%	Validation logs
Manual review scope	Majority of applications	High-risk only (~30%)	Routing analytics
Application approval rate	Not tracked	85%+	Decision engine reports
SLA compliance rate	Not tracked	95%+	SLA monitoring dashboard
Audit trail completeness	Paper-based	100% digital	Audit log coverage
KYC verification success rate	Manual / inconsistent	99%+	KYC API logs

12.10 Assumptions & Constraints

Assumptions:

- Customers have access to internet and a device
- Documents submitted are in PDF or image format
- Officers will be trained before system goes live
- Compliance rules are predefined and documented
- System assumes defined SLA thresholds for low, medium, and high-risk application processing
- External KYC API is assumed to be available under normal operating conditions, with retry and manual fallback mechanisms for failures

Constraints:

- Project is a portfolio simulation – no real development
- No actual third-party API integrations
- Timeline: 6 weeks
- Tools limited to SQL, Power BI, Draw.io, Figma, Excel, Word

13. Functional Requirements Document (FRD)

13.1 – FRD Overview

13.1.1 Purpose

This Functional Requirements Document defines the detailed functional specifications for the End-to-End Customer Onboarding System. It translates business requirements from the BRD into specific system behaviors, screen-level interactions, validation rules, decision logic, and exception handling to guide design and development.

13.1.2 Scope

This document defines the functional behavior, validation rules, decision logic, exception handling, and user interactions for all 13 modules of the Customer Onboarding System. It covers the complete onboarding lifecycle from customer registration through account activation.

13.1.3 System Modules

No.	Module	Description
1	Customer Portal	Registration, application form, document upload, status tracking
2	Document Upload Service	File handling, format checking, upload guidance
3	Document Validation Service	OCR processing, completeness check, validation rules
4	Data Extraction Engine	Auto-populate fields from validated documents
5	KYC Verification Service	External API integration, retry logic, fallback handling
6	Risk Scoring Engine	4-factor scoring model, classification, routing decision
7	Decision Routing Engine	Auto-approve, officer queue, compliance escalation
8	Officer Review Dashboard	Review queue, pre-filled data, SLA tracking, decision actions
9	Compliance Dashboard	High-risk escalations, enhanced due diligence, audit access
10	Notification Service	Email and SMS triggers at every key stage
11	Audit Logging Service	Immutable action logs, timestamps, actor tracking
12	Core Banking Integration	Automated account creation upon approval
13	Management Dashboard	Real-time KPIs, analytics, SLA monitoring, reporting

13.2 Functional Specifications per Module

13.2.1 Module 1 – Customer Portal

Field	Detail
Purpose	Provide customers with a digital self-service interface to register, complete application, upload documents, and track status without visiting a branch
Trigger	Customer visits onboarding portal URL from any device
Preconditions	Customer is not already registered with the same email or national ID
Inputs	First Name, Last Name, Date of Birth, Email, Phone Number, Residential Address, Document Type
Process	1. Customer visits portal → 2. Selects "Start Application" → 3. Completes form with real-time validation → 4. Uploads documents → 5. Reviews summary → 6. Submits → 7. Receives confirmation
Validations	Customer must be 18 years or older based on Date of Birth / Email must be unique / Phone must be valid format / All mandatory fields complete / Both documents uploaded before submission
Outputs	Application record created / Confirmation email and SMS sent / Application reference number generated
Postconditions	Application status set to "Submitted" / Document Validation Service triggered / Audit log entry created

Input Field Specifications

Field	Type	Max Length	Mandatory
First Name	Text	50 characters	Yes
Last Name	Text	50 characters	Yes
Date of Birth	Date	-	Yes
Email Address	Email format	-	Yes
Phone Number	Valid format	-	Yes
Residential Address	Text	-	Yes
Document Type	Dropdown	-	Yes

Error Handling

Error	Message Shown
Invalid email format	"Please enter a valid email address"
Email already registered	"An account with this email already exists. Please login or use a different email."
Customer underage	"You must be 18 or older to apply."
Mandatory field empty	Inline error shown on specific field
Session timeout	"Your session has expired. Your progress has been saved."

13.2.2 Module 2 – Document Upload Service

Field	Detail
Purpose	Handle secure document uploads with format and size validation before passing to Document Validation Service
Trigger	Customer reaches document upload step in application form
Preconditions	Customer has completed registration form successfully and has documents available
Inputs	Government-issued photo ID / Proof of address / File format: PDF, JPG, PNG / Max size: 5MB per document
Process	1. Display document checklist → 2. Customer selects document type → 3. Uploads file → 4. System checks format and size → 5. If valid → queue for validation / If invalid → show error
Validations	Format must be PDF, JPG, or PNG / Size must not exceed 5MB / Both documents required / Document must not be expired
Outputs	Files stored securely / Upload confirmation shown / Files passed to Document Validation Service
Postconditions	Upload status recorded in audit log / Validation Service triggered for each document

Error	Message Shown
Wrong format	"Please upload PDF, JPG, or PNG files only."
File too large	"File exceeds 5MB limit. Please compress and retry."
Missing document	"Both ID and proof of address are required before submitting."
Upload failed	"Upload failed. Please check your connection and try again."

13.2.3 Module 3 – Document Validation Service

Field	Detail
Purpose	Automatically validate uploaded documents for format, completeness, and readability using OCR technology
Trigger	Document Upload Service passes files to validation queue
Preconditions	Documents successfully uploaded and within accepted format and size limits
Inputs	Uploaded document files, document type metadata, application reference
Process	1. Receive documents → 2. Check format and size → 3. Run OCR → 4. Check completeness → 5. Check expiry date → 6. Return validation result

Validations	OCR must extract: name, DOB, ID number, expiry date / Document not expired / Image clear and readable / Document not issued within 30 days of application
Outputs	Validation result (Pass/Fail) / Extracted data passed to Data Extraction Engine / Failure reason sent to Notification Service
Postconditions	Validation result logged in audit trail / Data Extraction triggered if pass / Notification triggered if fail

Error	System Action
OCR failure	Document marked invalid — customer prompted to resubmit clearer image
Document expired	Validation failed — customer notified with reason
Document unreadable	Specific guidance returned to improve image quality
Recently issued document	Flagged as risk indicator — passed to Risk Scoring Engine with flag

13.2.4 Module 4 – Data Extraction Engine

Field	Detail
Purpose	Auto-populate customer record from OCR-extracted document data — eliminating manual data entry
Trigger	Document Validation Service returns successful validation result
Preconditions	Document validation passed / OCR extraction completed successfully
Inputs	OCR-extracted text fields from validated documents / Customer application reference
Process	1. Receive extracted data → 2. Map fields to system schema → 3. Cross-check against form data → 4. Flag mismatches → 5. Auto-populate customer record → 6. Pass to KYC Service
Validations	Extracted name must match form name / Extracted DOB must match form DOB / Extracted ID must match form ID / Address consistent with registered address
Outputs	Customer record auto-populated / Data consistency flags passed to Risk Scoring Engine / Mismatch flags recorded
Postconditions	Auto-population logged in audit trail / KYC Verification Service triggered / Mismatch flags stored for Risk Scoring input

Error Handling

Error	System Action
Extraction fails	Route to manual review queue with officer alert

Major data mismatch	Flag as Medium or High risk — passed to Risk Scoring Engine
Minor spelling variation	Flag as low-risk indicator — noted in risk summary

13.2.5 Module 5 – KYC Verification Service

Field	Detail
Purpose	Verify customer identity against government or third-party database via external API integration
Trigger	Data Extraction Engine completes auto-population successfully
Preconditions	Data extraction completed / External KYC API assumed available under normal conditions
Inputs	Customer name, DOB, ID number, document type, application reference
Process	1. Receive identity data → 2. Call external KYC API → 3. Receive response → 4. If success → pass to Risk Scoring / 5. If fail → retry x2 → 6. If still fails → manual review
Validations	Retry up to 2 times at 30-second intervals
Outputs	KYC result (Verified / Not Verified / Manual Review) / Result passed to Risk Scoring Engine
Postconditions	KYC result logged in audit trail with timestamp / Risk Scoring Engine triggered

API Response Handling

Response	System Action
Verified	Proceed to Risk Scoring Engine
Not Verified	Flag as High Risk — passed to Risk Scoring Engine
API Timeout	Retry after 30 seconds
API Failure x2	Route to manual review queue + officer alert

Error Handling

Error	System Action
API unavailable	Retry x2 → manual review if still failing
Timeout	Retry once after 30 seconds
Invalid response format	Log error → route to manual review
Manual review approved	Re-enter Risk Scoring Engine flow
Manual review rejected	Notification Service triggered — rejection sent

13.2.6 Module 6 – Risk Scoring Engine

Field	Detail
Purpose	Evaluate each application across 4 defined risk factors and produce a total risk score determining routing decision
Trigger	KYC Verification Service returns result
Preconditions	KYC verification completed or routed to manual review with result available
Inputs	KYC verification result / Data consistency flags / Duplicate detection results / External risk signal flags
Process	1. Receive all input signals → 2. Evaluate Factor 1 (Identity) → 3. Evaluate Factor 2 (Consistency) → 4. Evaluate Factor 3 (Duplicates) → 5. Evaluate Factor 4 (External Signals) → 6. Calculate total score → 7. Classify risk → 8. Pass to Decision Engine
Validations	Each factor capped at defined maximum / Total score cannot exceed 100 / Boundary scores: exactly 40 = Low / exactly 70 = Medium
Outputs	Total risk score (0–100) / Risk classification (Low/Medium/High) / Individual factor scores / Flagged risk indicators list
Postconditions	Score and classification logged in audit trail / Decision Routing Engine triggered

Scoring Factor Summary

Factor	Max Points	What It Checks
Factor 1 — Identity Verification	30	KYC API result, government database match
Factor 2 — Data Consistency	25	Cross-document data validation
Factor 3 — Duplicate Detection	25	Existing records, previous rejections
Factor 4 — External & Behavioral Signals	20	PEP, sanctions, geographic, behavioral flags

Error Handling

Error	System Action
Engine fails to load	Route to manual review + notify supervisor
Engine times out	Retry once after 30 seconds
Partial score only	Treat as High Risk by default
Score cannot be calculated	Flag application + notify supervisor immediately

13.2.7 Module 7 – Decision Routing Engine

Field	Detail

Purpose	Route each application to the correct decision path based on risk classification
Trigger	Risk Scoring Engine returns classification result
Preconditions	Risk scoring completed successfully
Inputs	Risk score / Risk classification / Application reference
Process	1. Receive classification → 2. Apply routing rule → 3. Move application to correct queue → 4. Trigger relevant dashboard update
Validations	Score 0–40 → Auto-approve / Score 41–70 → Officer queue / Score 71–100 → Compliance escalation
Outputs	Routing decision recorded / Application moved to correct queue
Postconditions	Routing decision logged in audit trail / Relevant dashboard updated / Customer notified of status update

Routing Decision Table

Score	Classification	Routing Action	SLA Target
0 – 40	Low Risk	Auto approved → Account creation triggered	Under 1 hour
41 – 70	Medium Risk	Officer Review Dashboard	Under 4 hours
71 – 100	High Risk	Compliance Dashboard — Enhanced Due Diligence	Under 24 hours

Error Handling

Error	System Action
Routing fails	Alert IT team + hold application in pending state
Invalid classification received	Default to High-Risk routing

13.2.8 Module 8 – Officer Review Dashboard

Field	Detail
Purpose	Provide onboarding officers a centralized review interface for medium-risk applications with full decision context
Trigger	Decision Routing Engine routes medium-risk application to officer queue
Preconditions	Application classified as Medium Risk / Officer logged in with valid credentials
Inputs	Pre-filled application data / Risk score and classification / Flagged risk indicators / Uploaded documents / Application timeline / SLA countdown
Process	1. Application appears in queue sorted by SLA urgency → 2. Officer opens application → 3. Reviews data, score, indicators, documents →

	4. Makes decision → 5. Enters reason → 6. Submits decision → 7. System processes outcome
Validations	Decision reason mandatory for rejection / Officer must always view all flagged indicators before deciding / SLA timer visible
Outputs	Decision recorded / Audit log entry created / Next step triggered (account creation or rejection notification)
Postconditions	Decision logged in immutable audit trail / Customer notified of outcome / SLA status updated

Officer Actions

Action	Next Step Triggered
Approve	Core Banking Integration — account creation
Reject	Notification Service — rejection with reason + resubmission link
Escalate	Compliance Dashboard — high-risk queue
Request Info	Notification Service — information request to customer

Error Handling

Error	System Action
SLA breached (4 hours)	Auto-escalate to senior officer + dashboard flag
Decision submitted without reason	"Decision reason is required."
System fails during submission	Decision held in pending — officer alerted to retry

13.2.9 Module 9 – Compliance Dashboard

Field	Detail
Purpose	Provide compliance team with escalated high-risk applications, full risk context, and enhanced due diligence tools
Trigger	Decision Routing Engine escalates high-risk application OR officer escalates medium-risk application
Preconditions	Application classified as High Risk or escalated by officer / Compliance officer logged in
Inputs	Full application data / Risk score (71–100) / All flagged risk indicators / Enhanced due diligence notes / Full audit trail
Process	1. Application appears in compliance queue → 2. Compliance officer reviews flagged indicators → 3. Reviews application history and audit trail → 4. Performs enhanced due diligence → 5. Makes decision → 6. Records detailed reason → 7. System processes outcome

Validations	Detailed reason mandatory for all compliance decisions / SLA: 24 hours maximum / All flagged indicators must be acknowledged before decision
Outputs	Decision recorded / Audit log entry created / Account creation or rejection triggered
Postconditions	Decision logged in immutable audit trail / Customer notified / Account created if approved or rejection sent if rejected

Error Handling

Error	System Action
SLA breached (24 hours)	Alert sent to management dashboard
Decision submitted without reason	"Detailed reason required for compliance decisions."
System fails during submission	Decision held in pending — compliance officer alerted

13.2.10 Module 10 – Notification Service

Field	Detail
Purpose	Send automated communications to customers and internal users at every key onboarding stage
Trigger	Any key system event (submission, validation result, decision, account creation, SLA breach)
Preconditions	Customer email and phone captured during registration / Event trigger received from relevant module
Inputs	Event type / Customer contact details / Application reference / Event-specific data (reason, link, status)
Process	1. Receive event trigger → 2. Identify notification template → 3. Populate with customer data → 4. Send via email and/or SMS → 5. Log notification in audit trail
Validations	All notifications must include application reference number / Rejection notifications must include specific reason and resubmission link
Outputs	Notification sent via email and/or SMS / Delivery status recorded
Postconditions	Notification logged in audit trail with timestamp and delivery status

Notification Templates

Event	Channel	Content
Application submitted	Email + SMS	"Application received. Reference: [ID]. We will update you shortly."

Document invalid	Email + SMS	"Document rejected. Reason: [X]. Please resubmit [link]"
KYC under review	Email	"Identity verification requires additional review. We will notify you within [SLA]."
Application approved	Email + SMS	"Your account is ready. Login here: [link]"
Application rejected	Email + SMS	"Application not approved. Reason: [X]. Resubmit within 7 days: [link]"
Resubmission reminder	Email	"Your application expires in 3 days. Please resubmit [link]"
Account confirmation	Email + SMS	"Account confirmed. Your login details: [details]"
SLA breach alert	Internal	"Application [ID] has breached SLA. Immediate action required."

Error Handling

Error	System Action
Email delivery fails	Retry once after 5 minutes
SMS delivery fails	Retry once — if still fails log and flag for manual follow-up
Invalid contact details	Flag application for officer review

13.2.11 Module 11 – Audit Logging Service

Field	Detail
Purpose	Record every system and user action with full traceability for compliance and regulatory audit purposes
Trigger	Any system event, user action, or decision at any stage
Preconditions	System event or user action has occurred / Logging service active
Inputs	Event type / Timestamp / Actor ID / Module name / Input data reference / Output result / Decision reason
Process	1. Receive event data → 2. Create log entry with all required fields → 3. Write to immutable audit log → 4. Confirm write success → 5. Generate backup if write fails
Validations	Every entry must contain all mandatory fields / Timestamps in standardized format / No entry can be edited or deleted after creation
Outputs	Permanent immutable log entry stored / Available for compliance audit retrieval
Postconditions	Log entry permanently stored / Accessible to authorized roles only

Access Rules

Role	Access Level

Compliance Team	Read only
Management	Read only
IT Team	Read only – system logs
Onboarding Officers	No access
Customers	No access

Error Handling

Error	System Action
Primary log write fails	Backup log generated + IT alert triggered
Logging service unavailable	System alert sent + temporary local log created

13.2.12 Module 12 – Core Banking Integration

Field	Detail
Purpose	Automatically create customer account in core banking system upon approval — eliminating manual IT intervention
Trigger	Approval decision received from Decision Routing Engine or Officer/Compliance decision
Preconditions	Application approved by relevant decision owner / Core banking system available
Inputs	Approved customer data (name, DOB, address, ID number) / Application reference / Account type
Process	1. Receive approval trigger → 2. Send customer data to Core Banking API → 3. API creates account → 4. Returns account number and credentials → 5. Pass details to Notification Service
Validations	All mandatory customer data fields present / Duplicate account check performed before creation
Outputs	Account number generated / Login credentials created / Account confirmation passed to Notification Service
Postconditions	Account creation logged in audit trail / Customer notified via Notification Service

Error Handling

Error	System Action
Core banking API unavailable	Retry after 5 minutes + IT alert
Duplicate account detected	Flag for manual review + officer alert
Account creation fails	Hold in pending state + IT team notified

13.2.13 Module 13 – Management Dashboard

Field	Detail
Purpose	Provide management with real-time operational visibility into onboarding performance, KPIs, and bottlenecks
Trigger	Dashboard accessed by management user OR automatic refresh every 60 seconds
Preconditions	Management user logged in with valid credentials / Data pipeline from all modules active
Inputs	Live application data from all modules / Historical records / SLA compliance data / Risk distribution data
Process	1. Aggregate data from all modules → 2. Calculate KPIs → 3. Display on dashboard → 4. Auto-refresh every 60 seconds → 5. Trigger SLA alerts if breached
Validations	Dashboard data must refresh within 60 seconds / SLA breach alerts must trigger within 5 minutes of breach
Outputs	Real-time KPI dashboard / SLA breach alerts / Compliance reports
Postconditions	Dashboard data refreshed every 60 seconds / SLA breach alerts triggered automatically

KPIs Displayed

KPI	Description
Total Applications	Count by status (Submitted / In Review / Approved / Rejected)
Approval Rate	% approved out of total completed
Average Processing Time	Mean time from submission to decision
Risk Distribution	% Low / Medium / High risk
SLA Compliance Rate	% completed within SLA target
Drop-off Rate	% abandoned before submission
Document Rejection Rate	% applications with invalid documents
Officer Performance	Average review time per officer

Error Handling

Error	System Action
Data refresh fails	Show last updated timestamp + retry alert
Module data unavailable	Show partial data with warning indicator

13.3 User Stories

13.3.1 Customer User Stories

ID	User Story	Priority
US-001	As a new customer, I want to register online so that I can open an account without visiting a branch	Must Have
US-002	As a new customer, I want real-time validation on my form so that I can fix errors before submitting	Must Have
US-003	As a new customer, I want a document checklist so that I know exactly what to upload	Must Have
US-004	As a new customer, I want to track my application status in real time so that I always know where my application stands	Must Have
US-005	As a new customer, I want instant notifications at every stage so that I am always informed without checking manually	Must Have
US-006	As a new customer, I want to resubmit documents via the portal so that I don't need to visit a branch if something is rejected	Must Have
US-007	As a new customer, I want to save my application progress so that I can complete it later if needed	Should Have

13.3.2 Onboarding Officer User Stories

ID	User Story	Priority
US-008	As an onboarding officer, I want a prioritized review queue so that I can focus on applications approaching their SLA deadline	Must Have
US-009	As an onboarding officer, I want pre-filled application data so that I don't have to manually re-enter information	Must Have
US-010	As an onboarding officer, I want to see the risk score and all flagged indicators so that I can make a fully informed decision	Must Have
US-011	As an onboarding officer, I want SLA countdown timers so that I know which cases need immediate attention	Must Have
US-012	As an onboarding officer, I want to approve, reject, or escalate applications directly from	Must Have

	the dashboard so that I can manage my queue efficiently	

13.3.3 Compliance Team User Stories

ID	User Story	Priority
US-013	As a compliance officer, I want to receive only high-risk escalations so that I can focus exclusively on complex cases	Must Have
US-014	As a compliance officer, I want to see all flagged risk indicators and enhanced due diligence notes so that I have full context before deciding	Must Have
US-015	As a compliance officer, I want access to the full immutable audit trail so that I can demonstrate regulatory compliance at any time	Must Have
US-016	As a compliance officer, I want automated SLA alerts so that no high-risk case exceeds the 24-hour limit	Must Have

13.3.4 Management User Stories

ID	User Story	Priority
US-017	As a manager, I want a real-time dashboard so that I can monitor onboarding performance without waiting for manual reports	Must Have
US-018	As a manager, I want approval rates by risk level so that I can identify process inefficiencies and take corrective action	Must Have
US-019	As a manager, I want SLA breach alerts so that I can intervene before deadlines are missed	Should Have
US-020	As a manager, I want to generate compliance reports so that I can share performance data with senior leadership	Should Have

13.4 Acceptance Criteria

The following acceptance criteria define the expected system behavior for each priority user story using Given / When / Then format.

13.4.1 US-001 – Online Registration

Field	Detail
Given	A new customer visits the onboarding portal
When	They complete and submit the registration form
Then	They must receive a confirmation email and SMS within 60 seconds containing their application reference number

13.4.2 US-002 – Real-Time Form Validation

Field	Detail
Given	A customer is filling the application form
When	They enter an invalid value (wrong email format, underage DOB, empty mandatory field)
Then	An inline error message must appear immediately on that specific field without page refresh

13.4.3 US-003 – Document Checklist

Field	Detail
Given	A customer reaches the document upload step
When	The upload page loads
Then	A checklist must display showing exactly which documents are required, accepted formats, and maximum file size before any upload begins

13.4.4 US-004 – Application Status Tracking

Field	Detail
Given	A customer has submitted their application
When	They log into the portal
Then	They must see their current application stage, last updated timestamp, and next expected action clearly displayed

13.4.5 US-005 – Real-Time Notifications

Field	Detail

Given	A customer has submitted their application and key events occur (document validation, decision, account creation)
When	Each key event is triggered in the system
Then	Customer must receive an email and/or SMS notification within 60 seconds of each event containing the application reference number and relevant status update

13.4.6 US-006 – Document Resubmission

Field	Detail
Given	A customer receives a document rejection notification
When	They click the resubmission link
Then	They must be able to upload corrected documents through the portal without visiting a branch and receive confirmation within 60 seconds

13.4.7 US-007 – Save Application Progress

Field	Detail
Given	A customer has partially completed their application
When	They close the browser or navigate away
Then	Their progress must be automatically saved and retrievable when they return to the portal

13.4.8 US-008 – Officer Review Queue

Field	Detail
Given	An onboarding officer logs into the dashboard
When	The queue loads
Then	Applications must be sorted by SLA urgency with risk level, risk score, and time remaining displayed for each application

13.4.9 US-009 – Pre-filled Application Data

Field	Detail
Given	An application is routed to officer review queue
When	Officer opens the application
Then	All customer data extracted from documents must be pre-filled and visible without requiring the officer to manually re-enter information

13.4.10 US-010 – Risk Score Visibility

Field	Detail
Given	An officer opens an application for review
When	The review screen loads
Then	The risk score, risk classification, and all flagged risk indicators must be visible before any approve or reject action is available

13.4.11 US-011 – SLA Countdown Timers

Field	Detail
Given	An application is sitting in the officer review queue
When	Officer views the queue
Then	A visible SLA countdown timer must be displayed for each application showing time remaining before breach

13.4.12 US-012 – Officer Decision Actions

Field	Detail
Given	An officer has reviewed an application
When	They select Approve, Reject, or Escalate
Then	The decision must be processed within 30 seconds, audit log entry created, and customer notified of outcome within 60 seconds

13.4.13 US-013 – Compliance Receives High-Risk Only

Field	Detail
Given	An application scores 71–100 on the risk engine
When	The Decision Routing Engine processes the result
Then	The application must automatically appear in the compliance queue within 60 seconds with full risk summary and flagged indicators

13.4.14 US-014 – Flagged Risk Indicators and Enhanced Due Diligence Notes

Field	Detail
Given	A compliance officer opens a high-risk application
When	The compliance review screen loads

Then	All flagged risk indicators, risk score breakdown by factor, and enhanced due diligence notes must be fully visible before any approve or reject action is available

13.4.15 US-015 – Full Immutable Audit Trail

Field	Detail
Given	A compliance officer needs to review the history of an application
When	They access the audit trail section for that application
Then	A complete chronological log must be displayed showing every action, timestamp, actor ID, and decision reason — with no ability to edit or delete any entry

13.4.16 US-016 – SLA Escalation

Field	Detail
Given	A high-risk application has been in compliance queue for 24 hours without a decision
When	The SLA timer expires
Then	The system must automatically send an alert to management and flag the application on admin dashboard within 5 minutes of breach

13.4.17 US-017 – Management Dashboard

Field	Detail
Given	A manager logs into the management dashboard
When	The page loads
Then	All KPIs including total applications, approval rate, average processing time, risk distribution, and SLA compliance must be visible and auto-refreshed within 60 seconds

13.4.18 US-018 – Approval Rates by Risk Level

Field	Detail
Given	A manager is viewing the management dashboard
When	They navigate to the approval rate section
Then	Approval and rejection rates must be displayed by risk level (Low/Medium/High) with the ability to filter by date range

13.4.19 US-019 – SLA Breach Alerts

Field	Detail
Given	An application has breached its SLA threshold
When	The SLA timer expires
Then	An automated alert must appear on the management dashboard and be sent via email to the relevant supervisor within 5 minutes of breach

13.4.20 US-020 – Compliance Reports

Field	Detail
Given	A manager needs to share performance data
When	They select "Generate Report" on the dashboard
Then	A downloadable compliance report must be generated showing approval rates, rejection rates, processing times, and risk distribution for the selected period

13.5 Use Cases

Use Case 1 – Customer Submits Application

Field	Detail
Use Case ID	UC-001
Actor	New Customer
Goal	Complete and submit onboarding application online
Precondition	Customer has internet access and required documents
Related FR	FR-001, FR-002, FR-003, FR-004
Business Rule Reference	Rule 1 – Document Validation (Trigger Point)
Main Flow	1. Customer visits portal → 2. Completes registration form → 3. Uploads documents via checklist → 4. Reviews summary → 5. Submits → 6. Receives confirmation with reference number
Alternative Flow	Document invalid → System sends alert → Customer resubmits via portal
Exception Flow	Customer fails resubmission 3 times → Application permanently closed → Customer notified
Postcondition	Application submitted, confirmation sent, Document Validation Service triggered

Use Case 2 – System Validates Documents

Field	Detail
Use Case ID	UC-002
Actor	System (Document Validation Service)
Goal	Automatically validate uploaded documents for format, completeness, and readability
Precondition	Customer has uploaded documents via Document Upload Service
Related FR	FR-005, FR-006, FR-007
Business Rule Reference	Rule 1 – Document Validation
Main Flow	1. Receive documents → 2. Check format and size → 3. Run OCR → 4. Check completeness and expiry → 5. Return validation result
Alternative Flow	OCR fails → Document marked invalid → Notification Service sends alert with specific reason
Exception Flow	Document unreadable → Validation fails → customer is notified to resubmit
Postcondition	Validation result logged — proceed to Data Extraction or notify customer

Use Case 3 – Risk Scoring and Routing

Field	Detail
Use Case ID	UC-003
Actor	System (Risk Scoring Engine + Decision Routing Engine)
Goal	Score application across 4 factors and route to correct decision path
Precondition	KYC verification completed or manual review approved
Related FR	FR-012, FR-013, FR-014, FR-015, FR-016, FR-017
Business Rule Reference	Rule 4 – Risk Scoring Logic
Main Flow	1. Receive KYC result and data flags → 2. Evaluate 4 factors → 3. Calculate total score → 4. Classify risk level → 5. Route to Decision Routing Engine
Alternative Flow	Score 0–40 → Application is Auto-approve Score 41–70 → Application is routed to Officer Review Queue Score71–100 → Application is escalated to Compliance Dashboard
Exception Flow	Engine fails → Manual review queue + supervisor alert
Postcondition	Application routed to correct decision owner — routing logged in audit trail

Use Case 4 – Officer Reviews Application

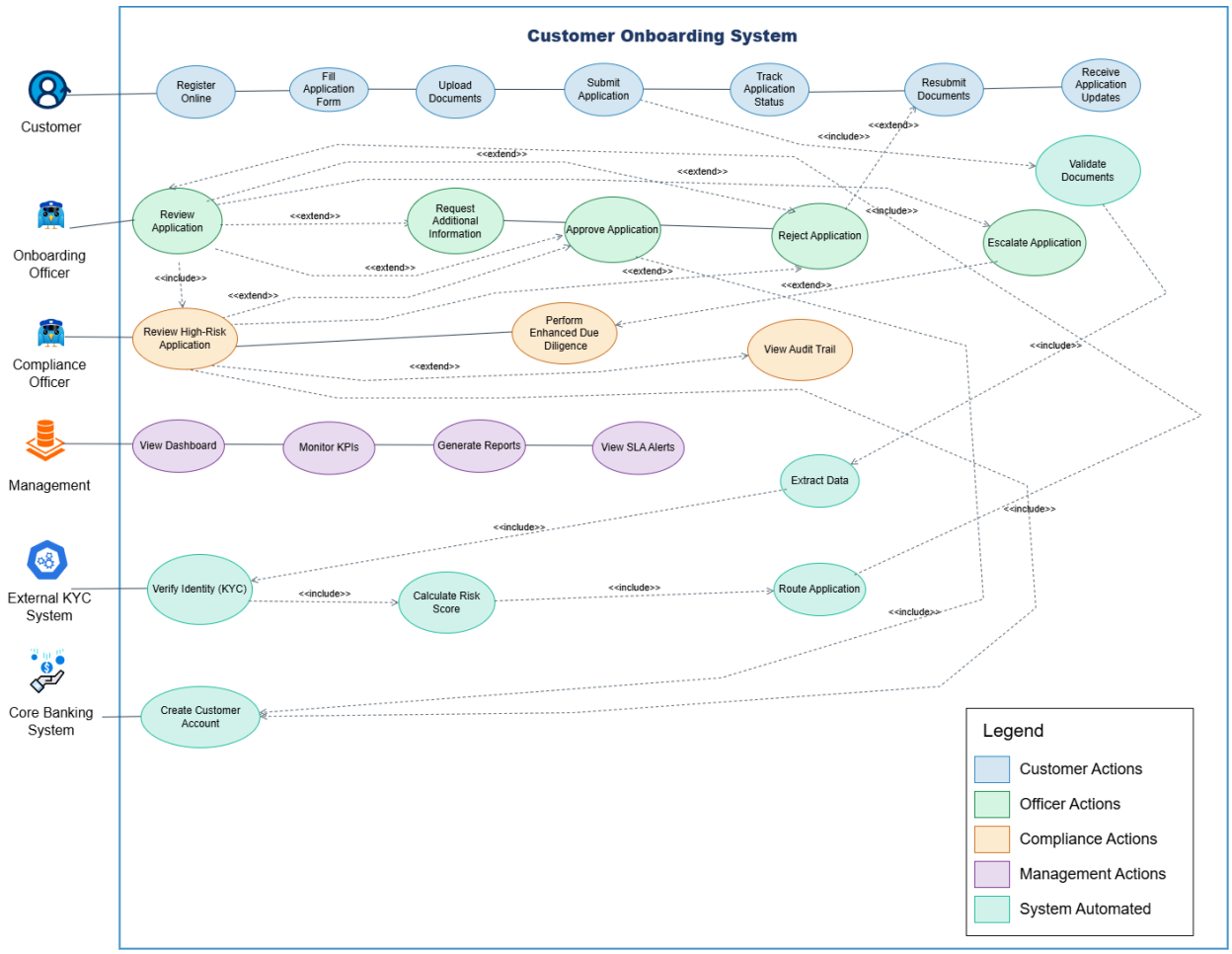
Field	Detail
Use Case ID	UC-004
Actor	Onboarding Officer
Goal	Review medium-risk application and make informed decision
Precondition	Application routed to officer queue with Medium Risk classification
Related FR	FR-016, FR-026, FR-028
Business Rule Reference	Rule 4 – Risk Scoring Logic (Medium Risk Routing)
Main Flow	1. Officer opens queue → 2. Selects application based on SLA urgency → 3. Reviews score, indicators, documents → 4. Makes decision → 5. Enters reason → 6. Submits decision
Alternative Flow	Officer escalates → Application routes to Compliance Dashboard
Exception Flow	SLA of 4 hours breached → Auto-escalate to senior officer + dashboard flag
Postcondition	Decision recorded in audit trail, customer notified of outcome, and next workflow step triggered

Use Case 5 – Compliance Reviews High-Risk Application

Field	Detail
Use Case ID	UC-005
Actor	Compliance Team
Goal	Review high-risk application and make compliance decision
Precondition	Application scored 71–100 and routed to compliance queue, or escalated by onboarding officer
Related FR	FR-017, FR-024, FR-025, FR-026, FR-029
Business Rule Reference	Rule 4 – Risk Scoring Logic
Main Flow	1. Open escalation queue → 2. Review flagged indicators and risk summary → 3. Review application history and audit trail → 4. Perform enhanced due diligence → 5. Make decision → 6. Record detailed reason → 7. System processes outcome
Alternative Flow	If approved → System triggers Core Banking Integration → Account is created automatically
Exception Flow	SLA of 24 hours breached → Alert sent to management dashboard

Postcondition	Decision logged in immutable audit trail and customer notified, with account created or rejection sent

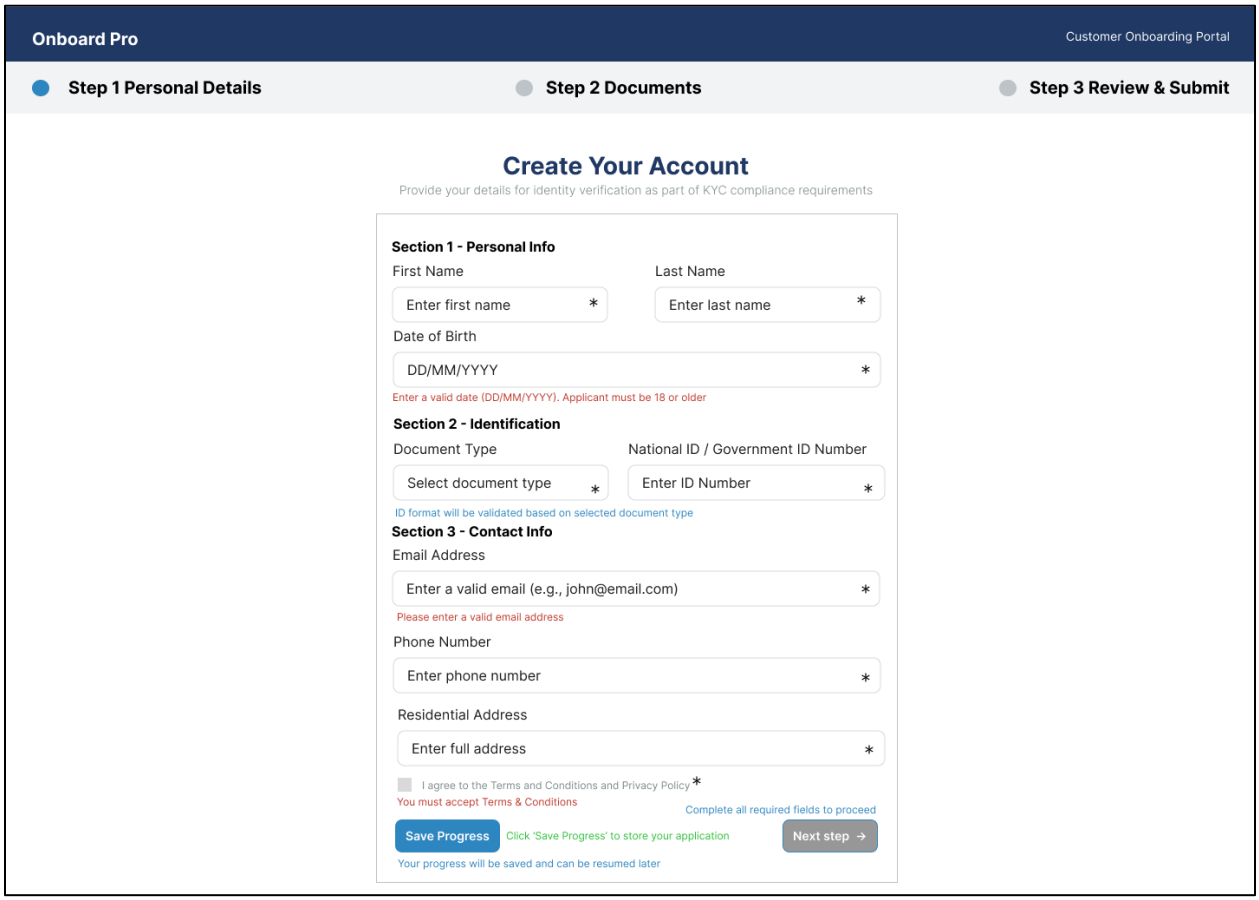
13.6 Use Case Diagram



14. Screen Design

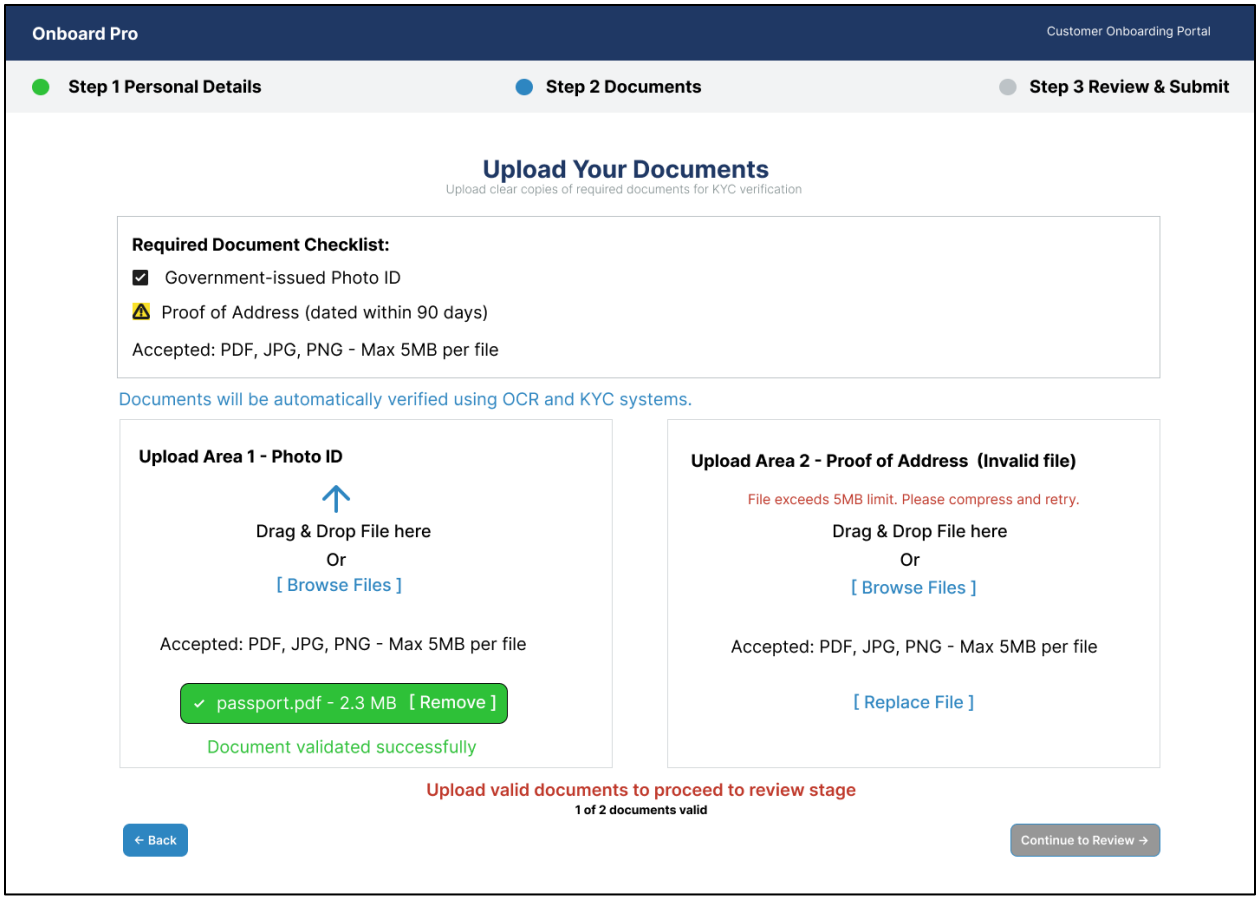
14.1 Customer Portal

This screen provides a digital self-service interface for customer registration and application initiation. It includes real-time validation for personal details such as name, date of birth, email, and ID information, ensuring data accuracy before submission.
It supports FR-001 and FR-002 by enabling online onboarding with inline validation and error handling.



14.2 Document Upload

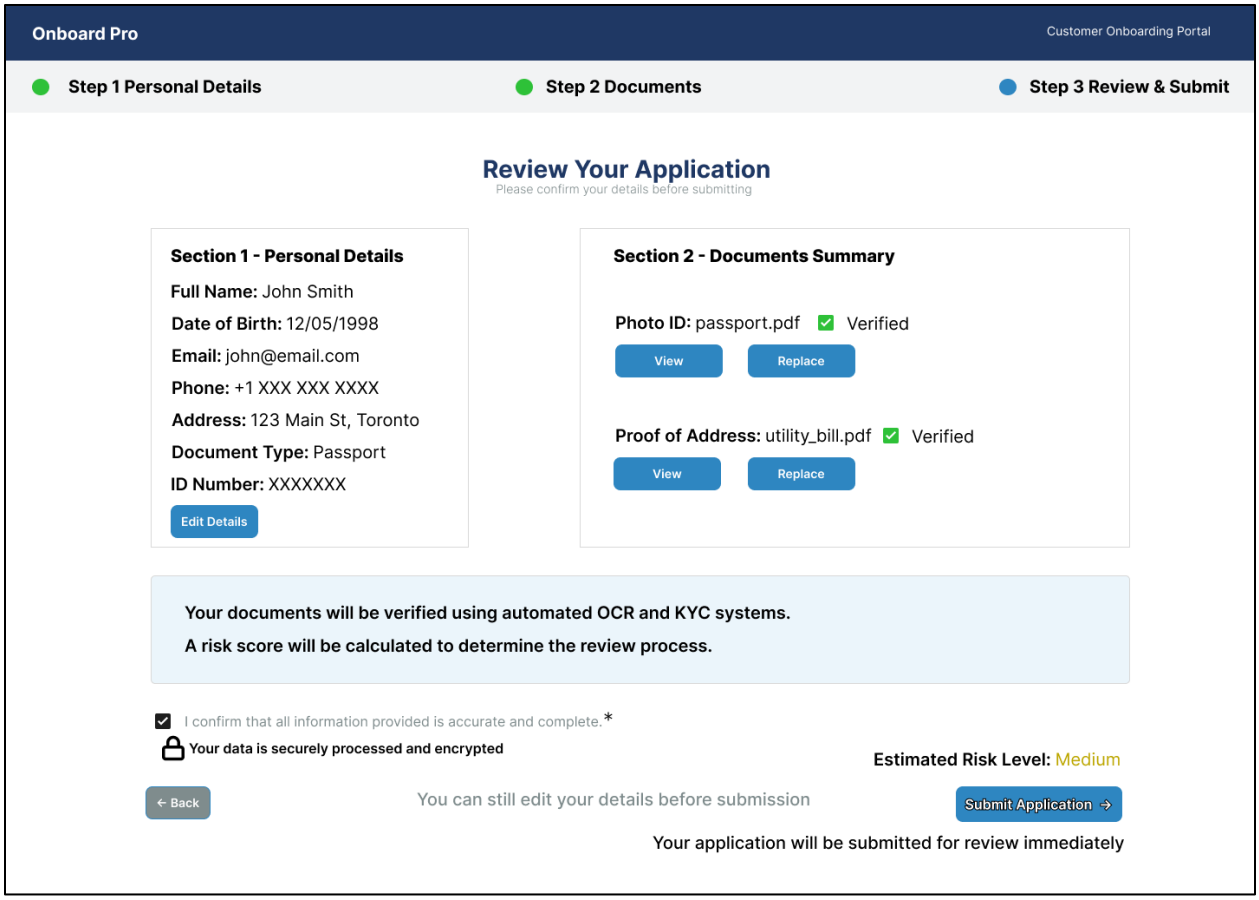
This screen allows customers to upload required documents with guided checklist validation. It enforces file format, size, and completeness rules before submission and provides immediate feedback on errors.
It aligns with FR-005 and FR-007, ensuring only valid documents proceed to the validation stage.



14.3 Review & Confirm

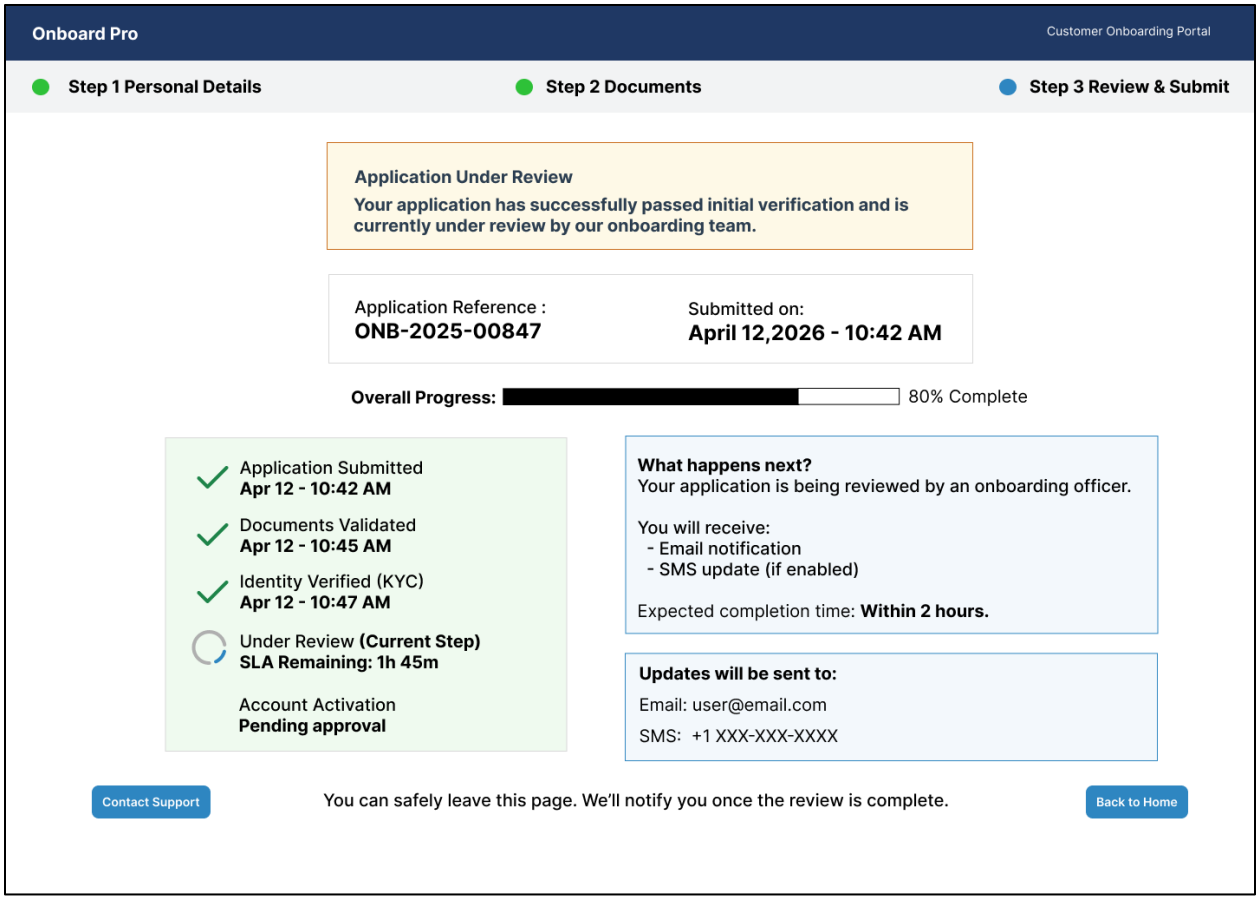
This screen enables customers to review all entered data and uploaded documents before final submission. It ensures confirmation of accuracy and provides transparency into the next steps, including risk scoring and verification.

It supports controlled submission and reduces data errors before processing.



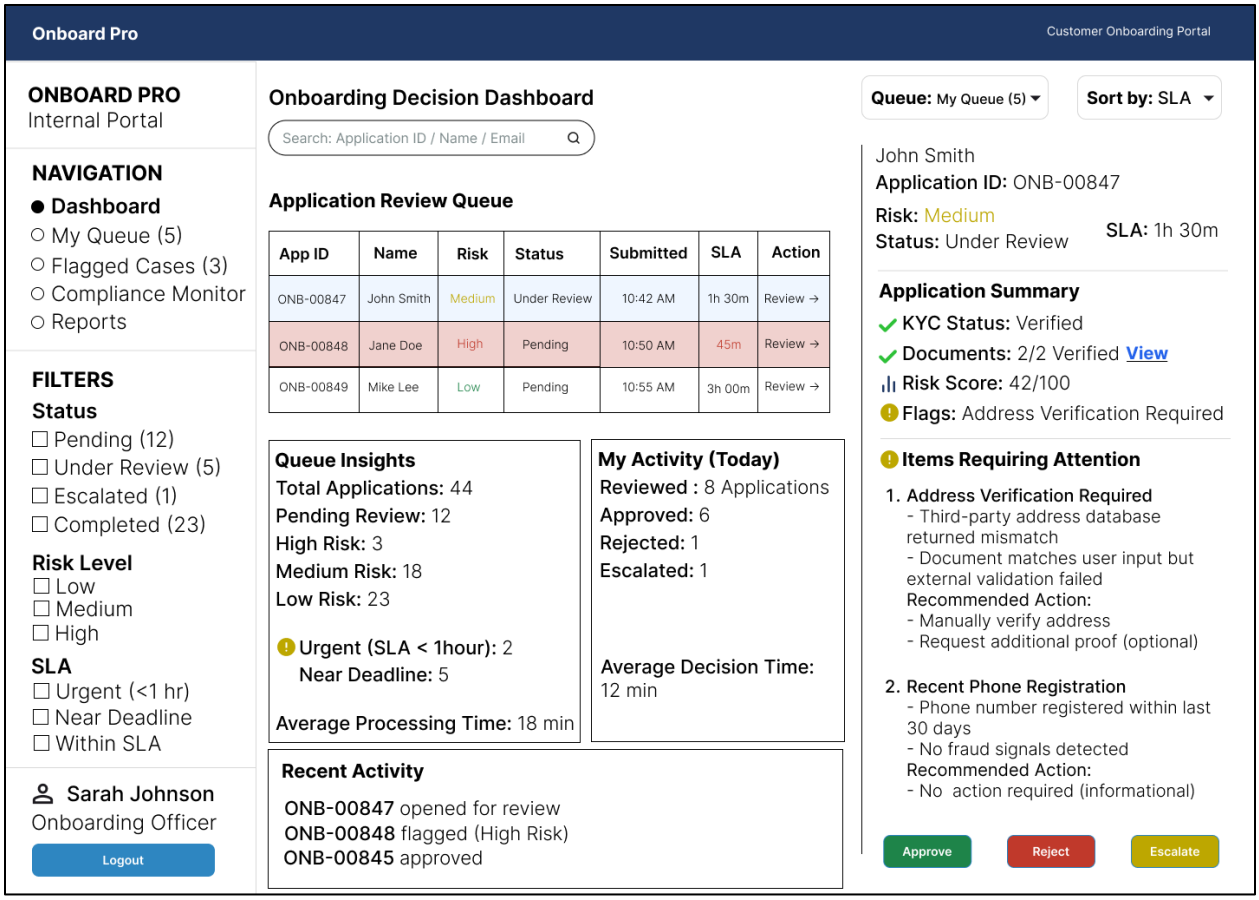
14.4 Application Status

This screen provides real-time visibility into application progress, including stages such as document validation, KYC verification, and review status.
As shown on page 1, it includes SLA tracking, progress percentage, and expected completion timelines, improving customer transparency and reducing support queries.



14.5 Onboarding Decision Dashboard

This internal dashboard allows onboarding officers to review medium-risk applications with full context, including risk score, flagged indicators, SLA timers, and application summary. It supports decision-making workflows (FR-016) and improves efficiency by prioritizing cases based on urgency and risk level.



14.6 Compliance Dashboard

This dashboard allows compliance officers to review high-risk applications (71–100) with full risk visibility, flagged indicators, enhanced due diligence tools, and complete audit trail access..

Onboard Pro Customer Onboarding Portal

ONBOARD PRO
Internal Portal

NAVIGATION
- ○ Dashboard
- ○ My Queue (5)
- ○ Flagged Cases (3)
- ● Compliance Monitor
- ○ Reports

FILTERS

Status
- ☐ Pending (12)
- ☐ Under Review (5)
- ☐ Escalated (1)
- ☐ Completed (23)

Risk Level
- ☐ Low
- ☐ Medium
- ☐ High

SLA
- ☐ Urgent (<1 hr)
- ☐ Near Deadline
- ☐ Within SLA

Sarah Johnson
Onboarding Officer
Logout

Compliance Monitoring Dashboard

Date: Today

Search: Application ID / Name / Email

System Overview

Total Applications: 120 ↑ +15%
Approved: 85 ↑ +3
Rejected: 20
Escalated: 15
High Risk: 23
Pending Review: 18
Average Processing Time: 18 minutes

SLA Status
- ✔ **Within SLA:** 95
- **Near Deadline:** 10
- ✘ **Overdue:** 5

Risk Distribution
Low Risk: 60
Medium Risk: 40
High Risk: 20

Risk Alert & Exceptions
- High Risk Spike detected in last 2 hours
- 5 applications breached SLA
- Repeated address verification failures
- Multiple applications from same phone number

Officer Performance (Today)
Sarah Johnson: 12 reviews (avg 10 min)
Mike Chen: 8 reviews (avg 14 min)
Alex Lee: 6 reviews (avg 18 min)

Fraud & Pattern Detection
Duplicate Applications: 3
Suspicious Phone Reuse: 2
Device Fingerprint Match: 1